



Agentic Imagery Analysis Over Time

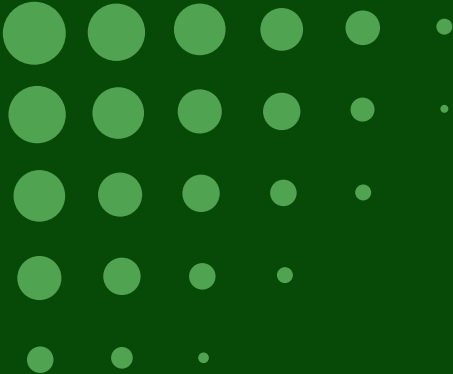
By Justin Rogers

Fall 2026 - GGS 662

Who Am I?

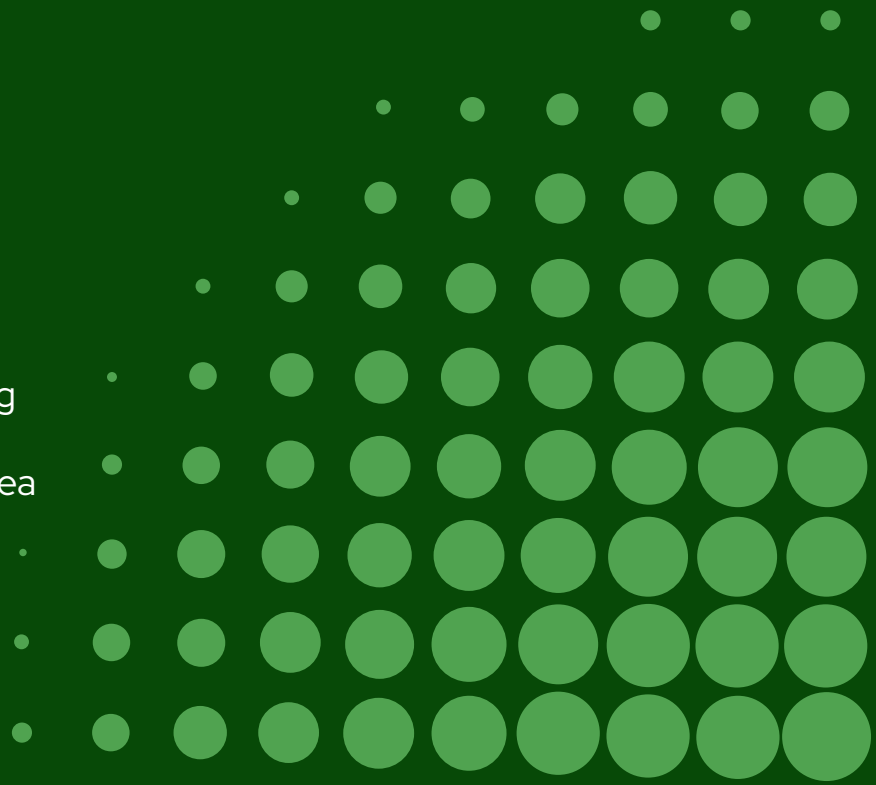
Justin Rogers

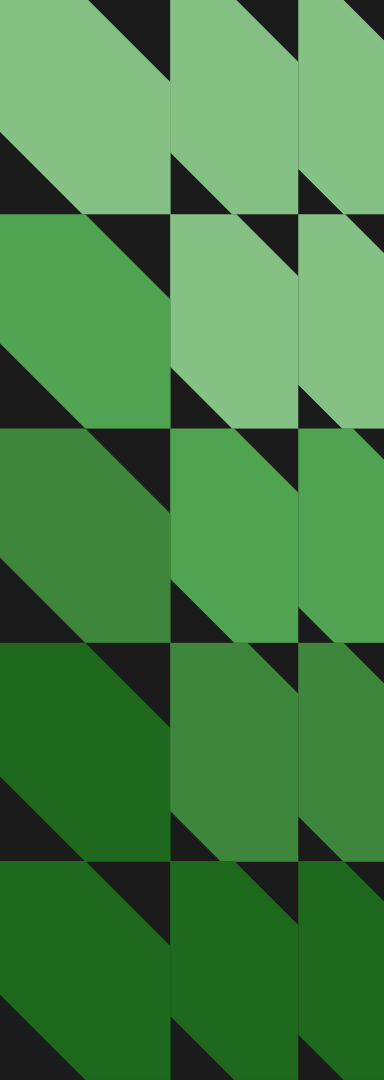
- Background: B.S Computational and Data Sciences, M.S. Computational Science
- Interests: AI and Computer Vision (CV) for Public Health and Safety
- Project Goals: Bring AI and Vision Language Models (VLMs) to GIS workflows
- Agent Goals: Enable AI Agents to improve analysis of satellite imagery



Research Question

How can AI agents enable analysts to better understand changes over time in massive imagery datasets, while using modern VLM technologies to detect natural and unnatural changes while textually describing findings regarding an area of interest?





Users & Inputs?

This project can potentially assist anyone monitoring an area of interest. For example, climatologists and investors of real estate can use this research improve monitoring techniques in an area of land for environmental health or potential value of investment.

Lowest level inputs include imagery of same area of interest, scaling up to only requiring areas of interest along with timeframes and potential data sources.



Tools & Technologies

Potential VLMs: Gemma VLM from Google, Phi-4 Vision from Microsoft, Llama from Meta

Data Sources: NASA Earthdata, USGS Earth Explorer, AWS Open Data

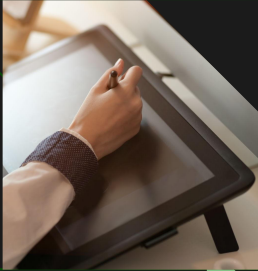
Coding Assistance: Claude Code

Human Validation

Outputs from VLMs should have image metadata and image markings for the detected changes across time. These changes should always be validated by a human in the loop since VLMs will never have a 100% accuracy.

Additionally, if changes are detected it is important to verify what the VLM is marking as different.

Outputs



01

Text Interpretation

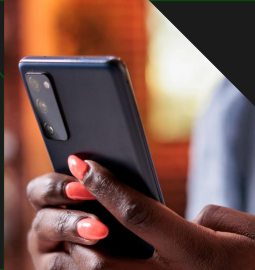
The Agentic VLM will attempt to describe the changes that it saw over time in the area of interest.



03

Confidence & Reasoning

To better understand how our model thinks and operates, the VLM should be using a reasoning model to report digital thoughts and confidence.



02

Visual Interpretation

The Agentic VLM should respond with provided imagery with outlines and markings that inspired conclusions.

Since VLMs are not perfect, we should be incorporating sub agents to validate and review the main agent's work to ensure detections and findings are accurate

Ability to execute

Image Agent

Second opinion on objects that have been detected as changing across images.

Geography Agent

Provides contextual knowledge based on a geographical region to better understand if what is going on makes sense in our area of interest.

News Agent

Searches the web for news articles in our area of interest to see if what is going on is making sense with current events in our area of interest

Weather Agent

Better understands the natural changes and anomalies in an area of interest over time.

Potential Sub Agents

How will we deal with bad data or bad responses

Bad Data

If our imagery is poorly collected, we should see this reflected in the confidence score of our VLM, this can also be confirmed by the image sub agent that is our second opinion of detections.

Bad VLM

Poor responses based on the information provided should be confirmed with various contextual agents or human interaction.

Hallucinations

All detections should be highlighted in the image remarks response by the VLM, allowing all visual hallucinations to be caught by the human in the loop validating these outputs.

Scope and Cost

1

Datasets are expensive, however with open data from commercial sources like NASA this cost can be lowered.

2

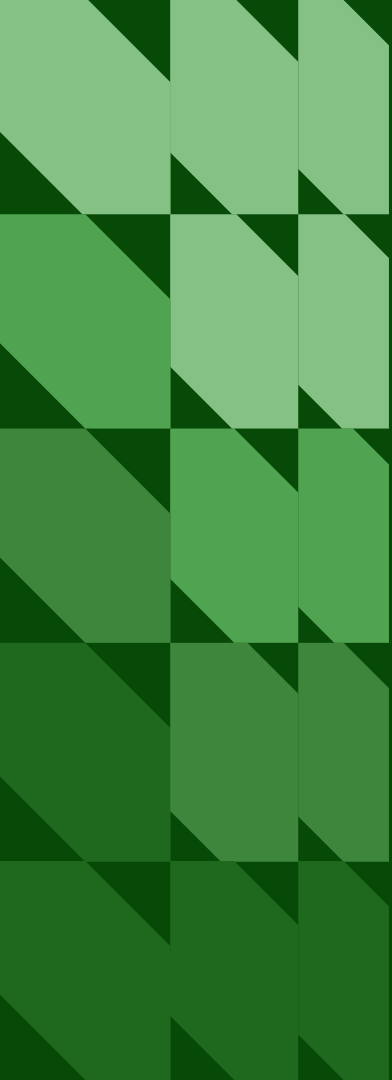
Open source VLMs make running models much more affordable rather than having to pay for vision tokens which can be rather expensive.

3

Additionally if the sub agents are using to many tokens, we can enable them only activate when confidence is getting low and rely on human validation other times.

4

Using university resources will allow for quick and efficient running of models as well as reduce and cloud computing costs if large models are required to increase accuracy



Thank you

Questions?